Machine Learning for teens

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Table of contents

Supervised vs. Unsupervised Learning

- **Supervised learning**:
  - You have input features **and** a known **target (label)**. The model learns to predict that target.
- **Unsupervised learning**:
  - You have input features **but no target**. The model tries to find patterns, structure, or groupings in the data on its own.

Types of supervised learning

- Regression
  - Once you have conations target value
- Classification
  - You have descript target value

Sigmoid function

- In mathematics, sigmoid function is function that convert any value to a value between 0 and 1
- Z is any number
- Y is a number between 0-1

$$y = \frac{1}{1 + e^{-z}}$$

Relationship between Sigmoid function and Logistic Regression

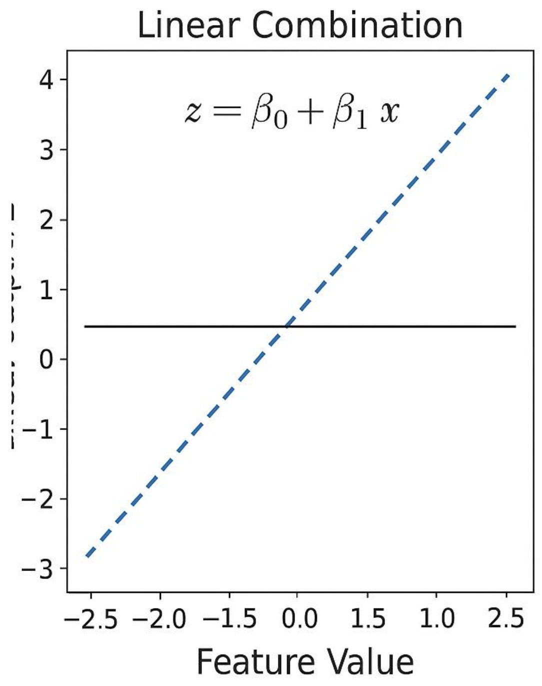
- In logistic regression, we first calculate a linear combination of the input features, then pass it through the sigmoid function to get a value between 0 and 1

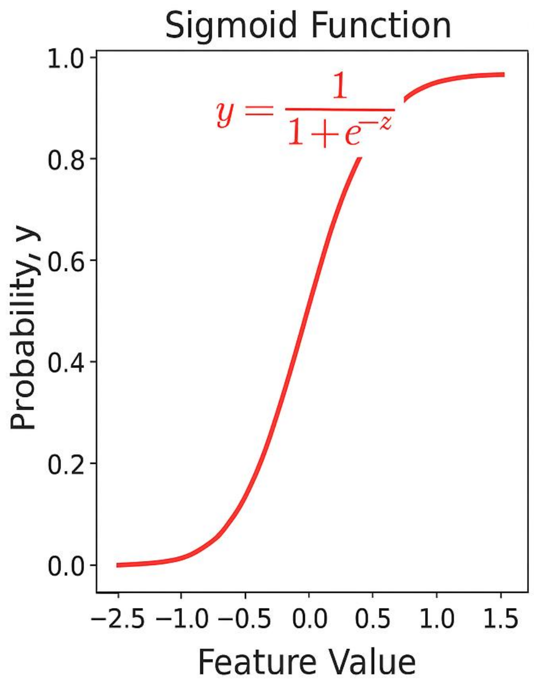
$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \Rightarrow y = \frac{1}{1 + e^{-z}}$$

linear regression sigmoid function

- Logistic regression is a supervised machine learning algorithm used for classification task

Relationship between Sigmoid Function and Logistic Regression





Manual Churn Prediction with Sigmoid

- **1. Create a Sample Dataset**
  - Age
  - Total_Purchase
  - Account_Manager
  - Years
  - Num_Sites
  - Churn (target variable)
- **2. Define the Sigmoid Function**
- **3. Assign Weights and Bias**
- **4. Compute Churn Probability Row-by-Row**
- **5. Classify Churn Based on Threshold**
- **For more detail see here**

Confusion Matrix

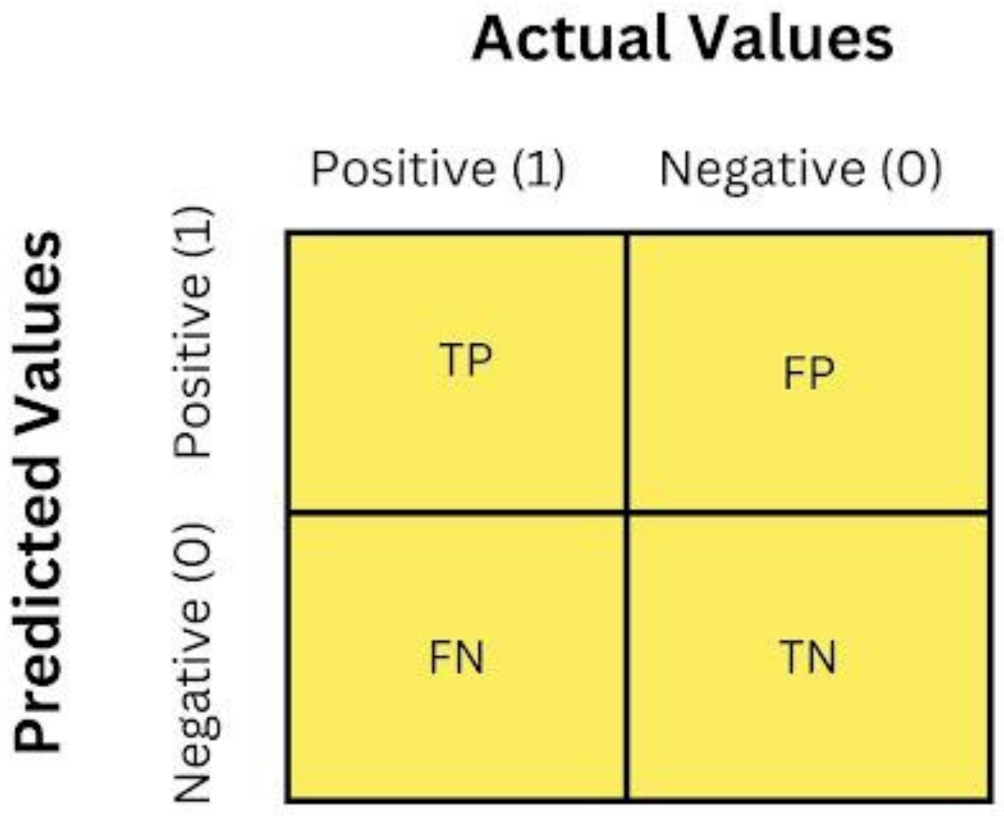
- A confusion matrix is a simple table that shows how well your model is performing by comparing its predictions to the actual results.
- Upload churn data from here
- Churn means
  - Churn = Yes → The customer left (canceled, unsubscribed, switched providers)
  - Churn = No → The customer stayed

Confusion Matrix

- The 4 Confusion Matrix Categories

Term	Meaning	Model Said	Reality
TP	Correct churn prediction	Churn	Churn
TN	Correct no-churn prediction	No churn	No churn
FP	Wrong churn prediction	Churn	No churn
FN	Missed churn	No churn	Churn

Confusion Matrix



key metrics

- Accuracy
  - How often is the model right?
- Precision
  - When the model says “churn,” how often is it correct?
- Recall
  - Out of all actual churners, how many did the model catch?
- F1-Score
  - A balance between precision and recall.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Scaling

- Scaling improves accuracy because it puts all features on the same scale, so the model treats them equally during training.
- See the end of the script where I applied scaling. If you look at the result, you'll see that scaling improves the outcome.

Assignment

- **Use the Titanic dataset and apply logistic regression to predict who survived.**